

Global Genomic Surveillance of Salmonella in the Environment: Assessing Virulence and Antimicrobial Resistance at Scale

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Legends to supplementary figures and tables

Fig. S1. Prevalence of antimicrobial resistance (AMR) genes across *Salmonella* subspecies

The heatmap illustrates the percentage frequency (0-100%) of various AMR genes identified in *S. enterica*, *S. diarizonae*, and *S. salamae*. The genes (rows) are categorized by their mechanism of resistance: antibiotic efflux, reduced permeability, antibiotic inactivation, and antibiotic target alteration. The colour gradient indicates prevalence, with purple representing high conservation (100%) and pink representing absence (0%).

Fig. S2. Comparative virulence profiling of *Salmonella* subspecies.

Heatmap displaying the percentage prevalence of virulence determinants across *S. enterica*, *S. diarizonae*, and *S. salamae*. The genes (rows) are categorized by their virulence mechanisms, and the colour gradient represents prevalence from low to high frequency across subspecies.

Table S1. Genomic metadata and accession identifiers for clinical reference genomes

Summary of the 12 *S. enterica* clinical isolates used as reference genomes in this study. The table lists serovar classification, specific strain nomenclature, and total genome length, along with corresponding NCBI BioProject and BioSample accession numbers.

Table S2. Genomic metadata, features, and assembly statistics for the *Salmonella* genome collection (n=1,399)

The table summarizes genomic metadata, sequence features, and assembly quality metrics for all isolates included in the study. Metadata includes isolate identifiers, source of isolation, and phylogenetic lineage/serogroup assignments. Genomic features encompass antimicrobial resistance (AMR) genes, virulence factors, and plasmid replicons identified using CARD, VFDB, and PlasmidFinder databases, respectively. Assembly statistics include genome size, contig number, N50 values, and other relevant quality metrics used for downstream comparative analyses. This dataset forms the basis for all downstream enrichment, phylogenetic, and ecological association analyses presented in the study.

Table S3. Distribution of antimicrobial resistance and virulence genes across plasmid replicons

The table summarizes the genetic features identified across the study isolates, including the contig accession numbers, gene names, and corresponding plasmid replicon types. Percent identity values are reported for both genes and plasmid replicons to indicate sequence similarity. Genes are functionally classified as AMR genes or virulence factors. This dataset provides the basis for identifying co-localization patterns and plasmid-mediated linkage between resistance and virulence determinants.

Table S4. Comparative prevalence and statistical enrichment of antimicrobial resistance, virulence, and plasmid-associated features across *Salmonella* serogroups

The table presents the distribution of genetic features identified using the CARD (antimicrobial resistance), VFDB (virulence factors), and PlasmidFinder databases. For each *Salmonella* serogroup,

the count and prevalence of specific features are compared against the rest of the dataset. Statistical significance of enrichment was assessed using P-values, and multiple-testing corrections were applied for adjusted P-values (Adj_P_Value).

Table S5. Statistical enrichment of genetic features, serogroups, and serovars across environmental and host-associated niches.

The table summarizes the differential distribution of genomic characteristics among surface water, soil, and host-associated environments. (A) Enrichment analysis of plasmid replicons (PL), virulence factors (VF), and antimicrobial resistance (AMR) genes, including prevalence across niches. (B) Distribution and statistical enrichment of *Salmonella* serogroups across environmental categories. (C) Niche-specific prevalence of *Salmonella* serovars. Statistical significance was assessed using P-values, with multiple-testing correction applied via the Benjamini-Hochberg false discovery rate (FDR; adjusted $P < 0.05$).